

# PROJECT PROPOSAL

## *De Novo Enzyme Design Using Geometric Graph Neural Networks and Deep Generative Diffusion Models*

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|---------------------------|----------------------------------------------|
| <b>Task Assignment:</b>   | TASK 4: Mini-Project / Proposal              |
| <b>Selected Track:</b>    | AI/ML Applications in Biotechnology Research |
| <b>Proposed Duration:</b> | 6 Months                                     |
| <b>Document Type:</b>     | Formal Academic Research Proposal            |

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## 1. INTRODUCTION

### 1.1 The Convergence of Artificial Intelligence and Biotechnology

The field of biotechnology is undergoing a foundational paradigm shift driven by the explosion of high-throughput biological data. Traditionally, discovering or engineering novel biomolecules relied heavily on empirical methodologies such as standard rational design or directed evolution. While highly effective, these approaches are inherently constrained by expensive trial-and-error laboratory cycles that demand significant physical resources, specialized biological reagents, and months of active execution. This creates a challenging bottleneck when rapid development is required to address emerging bio-manufacturing or therapeutic requirements.

Simultaneously, structural biology datasets have scaled exponentially over the past decade. Public repositories like the Protein Data Bank (PDB) and massive genomic archives now contain hundreds of thousands of highly resolved three-dimensional macromolecular coordinates and structural configurations. This structural abundance provides an ideal foundational environment for contemporary deep learning architectures. Rather than relying entirely on manual molecular modeling or static heuristics, modern Artificial Intelligence (AI) and Machine Learning (ML) frameworks can parse these high-dimensional geometric datasets to discover complex structural-functional relationships that are otherwise completely invisible to human biological intuition.

## 1.2 Generative AI in Biocatalysis

Enzymes serve as the essential biocatalysts required to drive green and sustainable chemical manufacturing, accelerate environmental bioremediation efforts, and synthesize high-value active pharmaceutical ingredients (APIs). However, wild-type or natural enzymes are rarely optimized out-of-the-box for rugged industrial deployment. They frequently denature, lose structural integrity, or drop precipitously in catalytic efficiency when exposed to non-physiological environments, such as elevated temperatures, organic co-solvents, or extreme pH levels.

This proposal outlines an end-to-end multi-disciplinary research framework focused on developing a generative artificial intelligence pipeline tailored specifically for *de novo* enzyme design. By treating primary amino acid arrangements as spatial geometric graphs and utilizing conditional continuous-time diffusion models, this project aims to design fully functional biocatalysts from scratch. This methodology allows scientists to bypass millions of years of natural evolutionary timeframes, establishing an automated pipeline to synthesize custom biological catalysts targeted at explicit chemical reactions.

## 2. PROBLEM STATEMENT

Despite recent breakthroughs in computational structural biology (such as forward folding networks), generating fully active, functional, and soluble synthetic proteins remains an unoptimized bottleneck within biotechnology. The primary challenges this project addresses include:

- 1. The Intractable Sequence Search Space:** For a relatively modest protein consisting of just 100 amino acids, the number of potential primary sequence permutations is exactly  $20^{100}$ . This astronomical figure represents a space that cannot be comprehensively sampled or scanned using conventional physical screening methods, manual mutagenesis, or brute-force algorithmic computing.
- 2. The Inverse Protein-Folding Challenge:** While platforms like AlphaFold have accurately solved the forward problem—predicting an established sequence's 3D structural fold—the inverse problem remains active. The field lacks robust computational tools capable of taking a target 3D active site geometry and accurately outputting an optimized, brand-new primary amino acid sequence that reliably folds back into that precise specification.
- 3. High Attrition Rates in Expression and Stability:** A significant majority of purely computational protein designs fail when converted *in vitro*. They often exhibit poor thermodynamic stability, aggregate into insoluble inclusion bodies inside common host chassis (such as *Escherichia coli*), or display a complete lack of real-world catalytic activity due to minute sub-angstrom structural distortions inside the active pocket.

Without an integrated, geometry-aware machine learning framework that implicitly binds structural physics with deep generative layers, the industrial discovery cycle for custom biocatalysts will remain expensive, slow, and highly unpredictable.

## 3. OBJECTIVES

The primary baseline objective of this mini-project is to design, implement, and computationally validate an end-to-end geometric deep learning pipeline that outputs novel, functional, and soluble enzyme sequences tailored to specific chemical transformations.

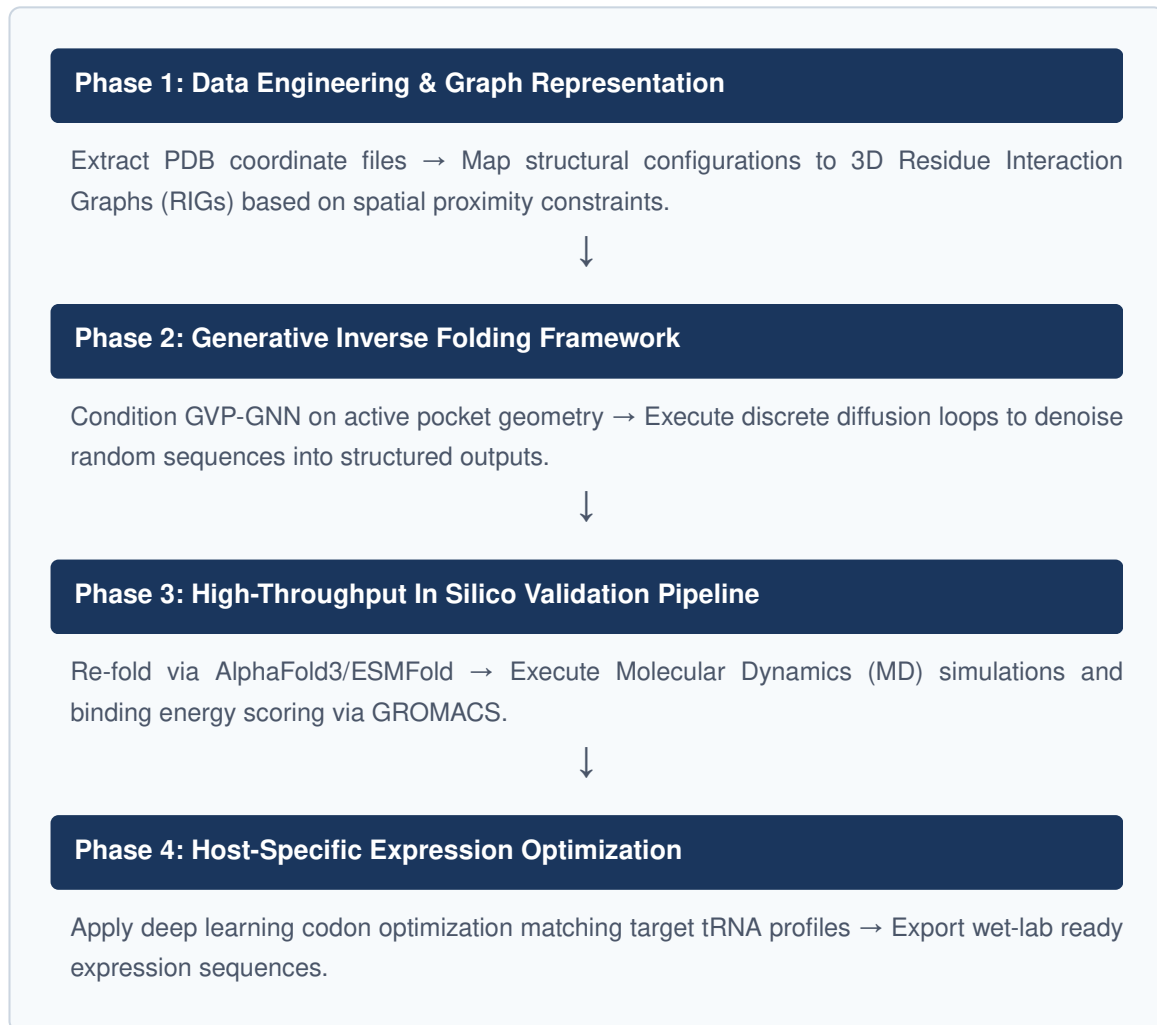
### Specific Research Objectives:

- **Objective 1:** Design and train a conditional **Geometric Vector Perceptron Graph Neural Network (GVP-GNN)** capable of mapping structural coordinates directly to categorical amino acid probabilities, thereby resolving the inverse folding problem for specified catalytic pockets.

- **Objective 2:** Build an automated, rigorous *in silico* validation dashboard that dynamically filters generated structural outputs based on calculated thermodynamic stability metrics ( $\Delta G$ ), target back-mapping alignment, and molecular docking scores.
- **Objective 3:** Establish a host-specific codon optimization module powered by language modeling to align generated structures with the exact tRNA abundance pools of standard biological expression host engines to maximize expression yields.

## 4. METHODOLOGY

The operational workflow of the proposed project bridges advanced machine learning architecture with strict biophysical filtering protocol, structured systematically across four operational phases.



### 4.1 Phase 1: Data Curation and Graphic Feature Engineering

The initial phase focuses on acquiring high-resolution crystallographic and cryogenic electron microscopy structural files (.pdb format) targeting major industrially relevant enzyme classes (such as hydrolases, transferases, and oxidoreductases) via the RCSB Protein Data Bank API. To preserve intrinsic three-dimensional geometric characteristics without relying on rotationally sensitive coordinate voxel grids, the proteins will be structurally represented as **3D Residue Interaction Graphs (RIGs)**. Within this mathematical abstraction, nodes designate individual amino acid residues (parameterized by their  $C_{\alpha}$  spatial positions, side-chain orientations, and localized chemical attributes), while directional edges map spatial proximity thresholds ( $\leq 8 \text{ \AA}$ ) and directional hydrogen-bonding networks.

## 4.2 Phase 2: Generative Model Architecture

The core computational engine utilizes an inverse-folding network driven by discrete generative diffusion models integrated directly over a GVP-GNN architecture. The model will be strictly conditioned on the spatial coordinates and structural dependencies of a defined target transition-state pocket. During generation, the diffusion model takes a noisy sequence matrix and iteratively denoises it into a distinct, high-probability primary sequence matching the geometric and electrostatic constraints of the target site. The network's structural training is governed directly by a multi-task optimization loss function balancing categorical cross-entropy of sequence alignment against total structural alignment error:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{sequence} + \beta \mathcal{L}_{structure}$$

## 4.3 Phase 3: Automated In Silico Validation Pipeline

To eliminate costly wet-lab failures, all machine-generated protein sequences will pass through a rigorous, automated multi-tiered computational screening filter. First, all sequence outputs will be folded *de novo* using modern structural models like *AlphaFold3* or *ESMFold*. Designs failing to achieve a Predicted Local Distance Difference Test (pLDDT) score  $\geq 85$  will be instantly culled. Second, surviving structures will undergo automated molecular docking via *AutoDock Vina* to evaluate binding free energies ( $\Delta G_{bind}$ ). Finally, short (50 ns) explicit-solvent Molecular Dynamics (MD) simulations will be executed inside *GROMACS* to confirm that the engineered catalytic pockets remain structurally stable under physical temperature fluctuations (310 K - 330 K).

## 4.4 Phase 4: Host Expression Optimization

The final layer processes validated amino acid sequences through a specialized deep learning translation engine. This module evaluates host-specific codon bias configurations, matching the generated sequences directly to the tRNA pool abundances of *Escherichia coli* BL21(DE3). This phase eliminates hazardous translational stalling, structural misfolding risks, and transcript instability, providing an optimized genetic sequence immediately ready for gene synthesis and physical wet-lab testing.

## 5. EXPECTED OUTCOMES

Over the course of the proposed 6-month timeline, this mini-project expects to successfully deliver actionable technical assets and demonstrate rigorous performance improvements relative to traditional protein engineering practices.

### 5.1 Technical Deliverables

- **Open-Source Production Pipeline:** A fully version-controlled, documented PyTorch code repository containing functional scripts for structural graph preprocessing, generative diffusion loops, and validation execution.
- **Validated De Novo Candidate Library:** A curated digital portfolio detailing 10–15 fully novel, non-natural synthetic enzyme sequences explicitly optimized to degrade complex environmental plastic polymers (e.g., polyethylene terephthalate variants) or stable industrial pollutants.

### 5.2 Quantitative Performance Evaluation Metrics

| EVALUATION METRIC CATEGORY    | BASELINE STANDARD (DIRECTED EVOLUTION)         | PROPOSED PROJECT TARGET BENCHMARK             |
|-------------------------------|------------------------------------------------|-----------------------------------------------|
| Design Generation Velocity    | Weeks to Months (Iterative Mutagenesis Cycles) | < 45 <i>ext{ Minutes}</i> per target sequence |
| Structural Folding Confidence | Highly Variable / Unpredictable Folds          | Average <i>ext{pLDDT Score}</i> ≥ 85.0        |
| Computational Viability Rate  | < 1% true positive structural fits             | ≥ 15% successful validation matches           |

## 6. REFERENCES

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